

DM



ED

PROJECT



GreenWave — RTOS-Based Emergency Vehicle Traffic Preemption

"Because an ambulance stuck in Cairo traffic is a clinical problem, not a traffic problem."

Team: Ziad Gaballah, Hamza Kamel, Khadeejah Iraky

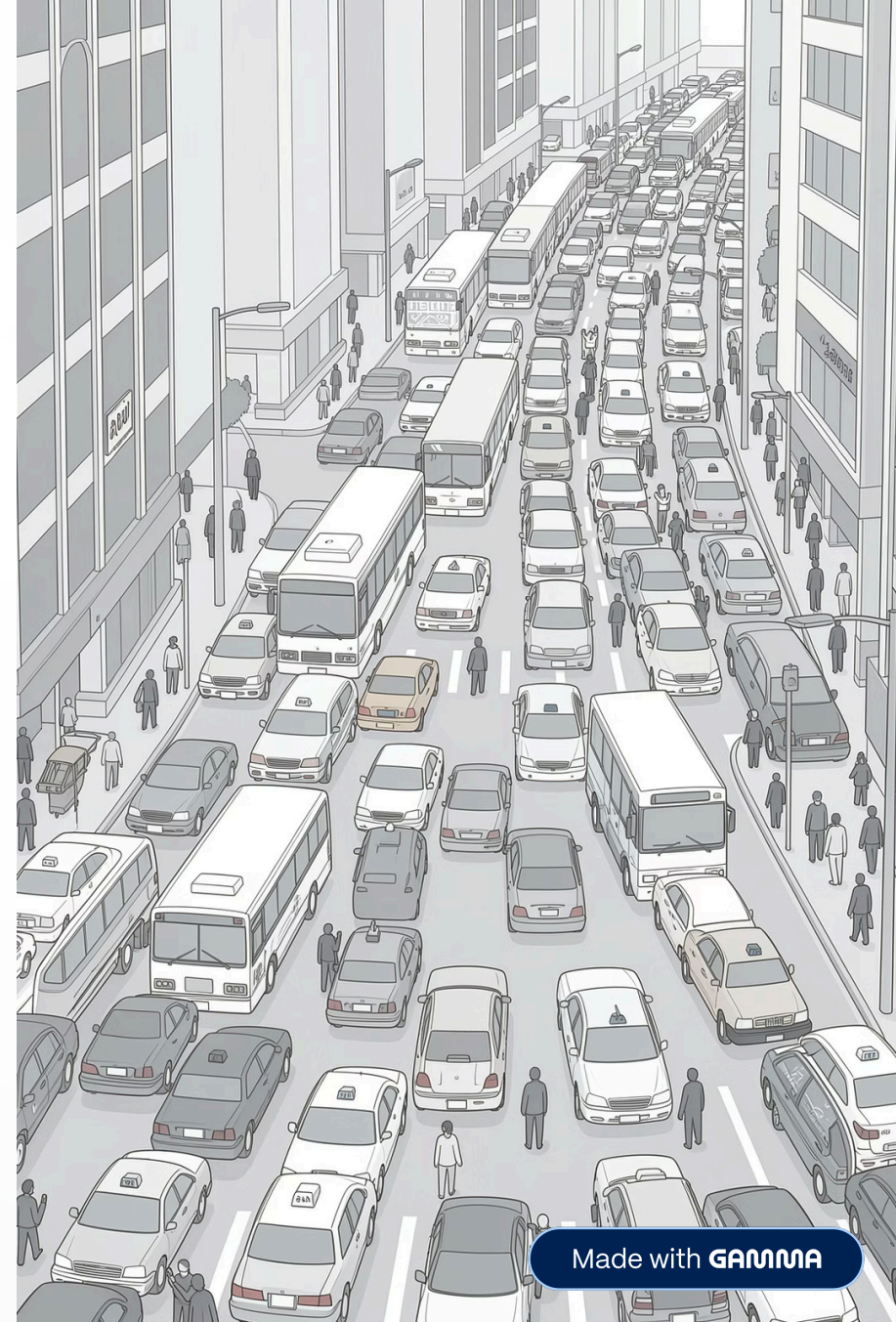
The Problem

Cairo ambulance response time: 8–15 minutes

National Ambulance Organization

Every minute of delay in cardiac arrest = ~10% drop in survival

No emergency vehicle preemption deployed anywhere in Egypt



This problem is already solved elsewhere. Just not here.

USA (Opticom)

15–25% faster response, deployed since the 1970s

Europe (Germany, UK, Ford V2X trials)

Lights auto-green + driver warnings

Gulf (Dubai, Abu Dhabi)

Smart ambulances + 100% signal coverage by 2026

Egypt

None deployed

 **Barrier in Egypt is institutional, not technical.**



Global Context: How EVP Works Elsewhere

Proven Systems

USA (Opticom): IR strobe + 900 MHz/2.4 GHz radio or cellular. 14–23% faster response. Deployed since 1970s. Competing stacks: EMTRAC, Econolite, TOMAR STROBECOM II.

Europe: Germany & UK use inductive-loop and IR on fire corridors. Current push: C-V2X / ITS-G5. Ford trialing C-V2X traffic lights that auto-green for emergency vehicles.

Gulf: Dubai rolling out UTC-UX Fusion across all major junctions by H1 2026 with 100% main-road coverage. UAE Ministry of Health launched smart ambulances that turn signals green and stream patient vitals to destination hospital.

Why Egypt Lags

- Cost of retrofitting legacy fixed-time controllers
- No V2X spectrum or standard allocated
- No procurement track record to force vendor competition
- Under-investment in ITS infrastructure overall

Why GPS + LoRa? Alternative Solutions Evaluated

Acoustic Siren Detection

Microphone arrays + spectral detection at intersections

- Cheap, retrofit-friendly
- But: noise-sensitive in dense traffic, unreliable

Computer Vision Detection

ML-based flashing light detection via intersection cameras

- Works in theory
- But: ML-heavy, scope inflator for this course

Cellular / MQTT Cloud Preemption

Pure cloud-based dispatch with SIM cards

- Works, proven
- But: adds broker complexity and cost for no pedagogical gain

Inductive-Loop / IR Legacy (Opticom-style)

Proven technology, pavement-embedded hardware

- Proven, mature
- But: requires pavement installation we cannot do

Our Solution: GreenWave



Ambulance broadcasts signed preemption request when in emergency mode

Intersection safely transitions to give the ambulance a green

Fail-safe: if anything goes wrong, all lights flash red

 **Two ESP32s. FreeRTOS on both. LoRa radio link.**

Architecture + Components

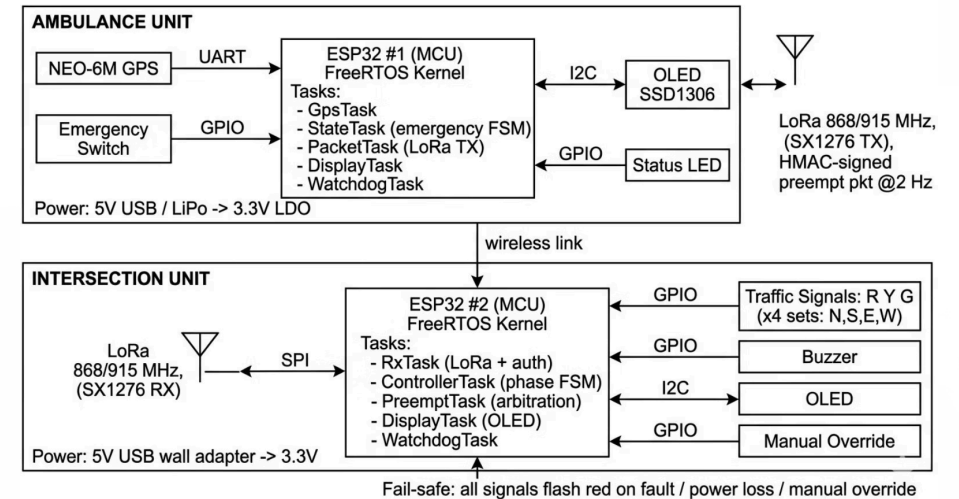
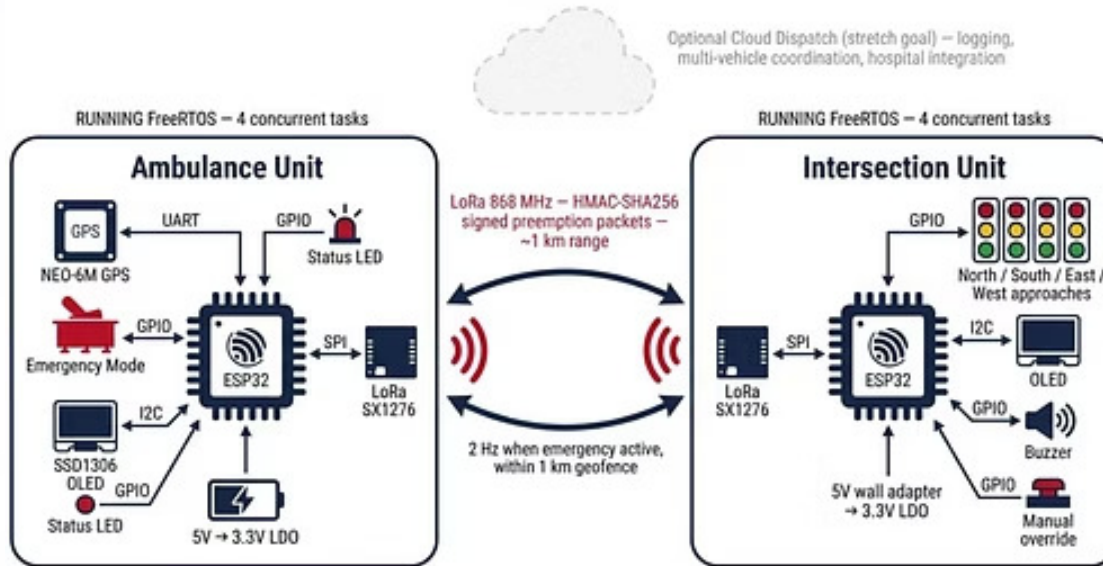
Ambulance Unit

- ESP32
- GPS (NEO-6M)
- Emergency Switch
- OLED
- LoRa SX1276

Intersection Unit

- ESP32
- 12 LEDs (4-way signal head)
- OLED
- Buzzer
- Manual Override
- LoRa SX1276

Hardware Block Diagram



Risks + Architecture Scope

Top Technical Risks

- GPS cold-start and indoor accuracy → test outdoors, dead-reckoning fallback
- LoRa range inside the lab → ESP-NOW as fallback
- Unsafe phase transitions → formal FSM + hardcoded clearances
- Integration slippage → subsystems working independently by M3

Scope Honesty

- "A real city deployment would be cloud-centralized."
- "We build the edge. Cloud dispatch is an optional stretch goal."
- "Safety-critical logic stays local to the intersection — by design."

Development Plan + Team



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Thank You!